

Evaluation of Restricted Diffusion in Cylinders. Phosphocreatine in Rabbit Leg Muscle

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Interpretation of NMR diffusion data is complicated in the presence of diffusion barriers. In this study, restricted diffusion in cylinders is evaluated. A method is presented to determine the diameter of the cylinder and the unrestricted diffusion coefficient from the dependence of the trace of the diffusion tensor with diffusion time. This method is valid even if the orientation of the cylinders is unknown and varies within the sample. An example is given for the diffusion of phosphocreatine in the cylindrically shaped fibers of rabbit skeletal muscle. © 1994 Academic Press, Inc.

INTRODUCTION

NMR diffusion spectroscopy is a useful technique to non-invasively determine the diffusional mobility. However, physical barriers to diffusion complicate the interpretation of experimental results. As a consequence, restrictions to the displacement by the boundaries of the restricting volume must be taken into consideration (1–3). For cylindrical volumes, the analysis method must also account for the orientation within the volume of interest. In addition, the orientation of an ensemble of cylinders may not even be similar within this volume. This situation is encountered *in vivo*, for example, in muscle fibers and axons. Here we present a new method to determine the unrestricted diffusion coefficient and the cylinder diameter based on the dependence of the trace of the diffusion tensor on diffusion time.

The method is applied to the diffusion of phosphocreatine in muscle cell. Restriction and orientation effects are demonstrated. The unrestricted diffusion coefficient and fiber diameter are determined assuming cylindrically shaped muscle fibers. The calculated diameter is in agreement with known muscle fiber diameter. Preliminary data have been published previously (4, 5).

THEORY

Diffusion

Diffusional motion in an unbounded and homogeneous medium is given by This is called GAUSSIAN diffusion

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$$P(\Delta \mathbf{r}) = \frac{(\Delta \mathbf{r})^2 / 4 D t_{\text{dif}}}{\sqrt{4 \pi D t_{\text{dif}}}} \quad [1]$$

which describes the probability P of displacement $\Delta \mathbf{r}$ in a diffusion time t_{dif} , where D is the diffusion coefficient (6). The root-mean-square (rms) displacement (λ) in one direction follows directly from this distribution:

$$\lambda_x^2 \equiv \langle (\Delta x)^2 \rangle = \frac{\int_V \int_V (x - x')^2 P(x - x') d\mathbf{r} d\mathbf{r}'}{\int_V \int_V d\mathbf{r} d\mathbf{r}'} = 2 D t_{\text{dif}}. \quad [2]$$

In fact, this can be taken as the definition of the diffusion coefficient. Especially in diffusion-sensitive NMR experiments, as suggested by Stejskal and Tanner (7), this definition is useful, as one actually measures the displacement. From Eq. [2], it is clear that the displacement will increase with longer diffusion time.

Restriction

In a finite volume, however, the displacement is limited by the boundaries of the volume. For infinitely long diffusion times, the displacement will be determined by the dimensions of the volume, whereas the influence of the diffusion coefficient of the unbounded medium is no longer important. This effect of the volume boundaries is called restriction. Equation [2] also shows that the diffusion coefficient will decrease for increasing diffusion times in the case of restricted diffusion. The exact dependence of the rms displacement on the diffusion time is determined by the shape of the limiting volume and the diffusion constant for the unbounded medium. Measurement of the displacement can therefore reveal information about the shape, or at least the dimensions of the volume.

In the limit of infinitely long diffusion times, the rms displacement follows directly from integration of the quadratic distance between all points within the volume, as any particle

starting at a position $\mathbf{r}$ will have equal probability to end at any position $\mathbf{r}'$. For a cylindrical volume with radius R , this leads to

$$\lambda_{\perp}^2 = \frac{\int_0^{2\pi} \int_0^R \int_0^{2\pi} \int_0^R [r \cos(\theta) - r' \cos(\theta')]^2 r r' dr d\theta dr' d\theta'}{\pi^2 R^4}$$

$$= \frac{1}{2} R^2 \quad [3]$$

for the displacement perpendicular to the symmetry axis of the volume. The displacement in the parallel direction is of course unrestricted. The effects of restriction can be accounted for by applying the appropriate boundary condition for the diffusional motions or can be incorporated in the diffusion coefficient.

Anisotropy

The restriction will lead to a different diffusion coefficient for different directions of measurement; the diffusion becomes anisotropic. In general, this is described by a tensor of rank two, i.e., a three-by-three matrix. By symmetry, most of the elements of this tensor are zero in the cell frame of reference, leaving only the diagonal elements. Assuming cylindrical symmetry, one can write the diffusion tensor as

$$\mathbf{D} = \begin{bmatrix} D_{\perp} & 0 & 0 \\ 0 & D_{\perp} & 0 \\ 0 & 0 & D_{\parallel} \end{bmatrix}. \quad [4]$$

There are only two independent elements, the diffusion perpendicular to the cylinder axis ($D_{\perp}$) and the diffusion parallel to it ($D_{\parallel}$). In a NMR diffusion experiment, elements of this tensor are probed by applying gradients, which results in a term $\mathbf{g} \cdot \mathbf{D} \cdot \mathbf{g}$ in the generalized Bloch equations (7, 8):

$$\frac{\partial \mathbf{M}}{\partial t} = \gamma \mathbf{M} \times \mathbf{B} - \frac{(M_x \hat{x} + M_y \hat{y})}{T_2} + \frac{(M_0 - M_z) \hat{z}}{T_1} + \nabla \cdot \mathbf{D} \cdot \nabla \mathbf{M}. \quad [5]$$

The direction in which the diffusion is measured is determined by the gradient ($\mathbf{g}$). The general solution of this equation is

$$\ln(A) = \ln(A_0) - \gamma^2 \int_0^{t'} \left[\int_0^{t''} \mathbf{g} dt'' \right] \times \mathbf{D} \cdot \left[\int_0^{t'} \mathbf{g} dt'' \right] dt', \quad [6]$$

where A_0 is the signal amplitude without diffusion, and $\mathbf{g}$ is the gradient function. Refocusing pulses in the experiment are incorporated by an inversion of the gradient function in the period prior to the pulse. The solution is valid only on top of the echo (or echoes), that is, the integral over $\mathbf{g}$ must be zero. Also, it is valid only for unrestricted diffusion, or restriction with infinitely short gradient pulses.

From the form of the diffusion term in Eq. [6] ($\mathbf{g} \cdot \mathbf{D} \cdot \mathbf{g}$), it is clear that one measures a diagonal element of a diffusion tensor in the gradient frame of reference. Note that this is not necessarily the laboratory frame, as the gradient can be a combination of one or more physical directions (g_x, g_y, g_z). The resulting diffusion coefficient is linked to the cell-frame diffusion tensor by a rotation

$$\mathbf{D}' = \mathbf{R}^{-1} \cdot \mathbf{D} \cdot \mathbf{R}, \quad [7]$$

where $\mathbf{D}'$ is the diffusion tensor in the gradient frame of reference, $\mathbf{D}$ is the diffusion in the cell frame, and $\mathbf{R}$ is the rotation transforming one frame to the other.

In order to calculate the size of a compartment in a certain direction from the restricted displacement, the diffusion in that particular direction must be selected, e.g., $D_{\perp}$ for the measurement of the cell diameter. This requires a diffusion gradient in exactly the same direction and thus prior knowledge of the orientation of the sample. In *in vivo* experiments this is often not available, and the applied gradient may therefore not be aligned with the cell axis. The resulting diffusion in a mixture of the different elements of the diffusion tensor, reflecting the different restriction sizes, as described by Eq. [7]. For a gradient with angle θ with respect to the principal cell axis, this would result in

$$D = \cos^2(\theta) D_{\parallel} + \sin^2(\theta) D_{\perp}. \quad [8]$$

This formula can also be derived from the weighted average rms displacement λ along the different axis. This mixing of diffusion coefficients makes it impossible to estimate the cell diameter by measuring only a single direction of diffusion.

Trace of Diffusion Tensor

One solution to the problem of the orientational effects is to acquire the diffusion in six different directions [9], which results in all the independent elements of the diffusion tensor in the laboratory frame. This leads to the information of the different cell-frame diffusion coefficients, as well as the rotations involved. However, in *in vivo* experiments, the situation is often even more complicated, as the orientation of the cells within the sample is not necessarily uniform. The effective $\mathbf{D}'$ is then an average of the different angles in the (selected) volume of interest. To solve this dependence on the orientation, we propose to measure the diffusion in just three orthogonal directions and then add the three resulting

diagonal elements (in the laboratory frame of reference), i.e., calculate the trace of the diffusion tensor. This is equal to the trace of the cell-frame diffusion tensor,

$$\begin{aligned} \frac{1}{3} \text{Tr}(\mathbf{D}') &= \frac{1}{3} (D'_{11} + D'_{22} + D'_{33}) \\ &= \frac{1}{3} (D_{||} + 2D_{\perp}) = \frac{1}{3} \text{Tr}(\mathbf{D}) = D_{av}, \end{aligned} \quad [9]$$

since the trace of a matrix is invariant under rotations. This operation removes any dependence on orientation. The dependence of the diffusion on the diffusion time results in the cell diameter. By substitution of the rms displacement in Eq. [9], the trace of the diffusion tensor can be expressed as

$$2t_{\text{dif}} \text{Tr}(\mathbf{D}) = \lambda_{||}^2 + 2\lambda_{\perp}^2 = \lambda_t^2, \quad [10]$$

where λ_t^2 is the 3D mean-square displacement. As can be seen from Eq. [10], the trace reflects the rms displacement in three dimensions [$\langle(\Delta \mathbf{r})^2\rangle$].

Cylindrical Restriction

The displacement in the restricted direction, is, as mentioned above, determined by the shape of the volume and the “free” diffusion coefficient (in this case $D_{||}$) together with the diffusion time. To calculate the diffusional attenuation in a NMR experiment, the displacement during the gradients also should be taken into account. A solution for this problem is the theory presented by Neuman (10). This is based on calculation of the mean-square phase effect produced by the displacement and the gradients, with the assumption that the phase distribution continues to be Gaussian. For diffusion perpendicular to the cylinder axis, the classical Stejskal and Tanner diffusion experiment, with two gradient pulses of amplitude g , duration δ , and separation Δ , results in

Signal attenuation for intra-axonal radial diffusion

$$\ln(A) = \ln(A_0) - 2\gamma^2 g^2 \sum_{m=1}^{\infty} \frac{[2D_f \alpha_m^2 \delta - 2 + 2e^{-D_f \alpha_m^2 \delta} + 2e^{-D_f \alpha_m^2 \Delta} - e^{-D_f \alpha_m^2 (\Delta - \delta)} - e^{-D_f \alpha_m^2 (\Delta + \delta)}]}{D_f^2 \alpha_m^6 (R^2 \alpha_m^2 - 1)}, \quad [11]$$

where D_f is the unbounded (free) diffusion constant, R is the radius of the cylinder, and α_m are the roots of the equation

$$J'_1(\alpha_m R) = 0. \quad [12]$$

J'_1 is the derivative of the Bessel function of the first kind, order one.

It can easily be shown that in the narrow pulse approximation ($\delta \ll \tau$), the diffusion attenuation is

$$\ln(A) = \ln(A_0) - 2\gamma^2 g^2 \delta^2 \lambda^2. \quad [13]$$

For long diffusion times and short gradient pulses, Eq. [12] approaches

$$\ln(A) = \ln(A_0) - \gamma^2 g^2 \delta^2 R^2, \quad [14]$$

corresponding to Eq. [13] with λ from Eq. [3].

EXPERIMENTS AND METHODS

For the acquisition of ^{31}P NMR spectra, a stimulated-echo sequence was used (Fig. 1) with the addition of the diffusion gradients (11). The application of this sequence has two advantages: the localization of a volume is easily achieved by the combination of three selective pulses with three different gradients, and, during the TM period, the magnetization is longitudinal. The latter is important since, under *in vivo* conditions, the longitudinal relaxation is much slower than the transverse relaxation (T_2). The localized volume was $6 \times 6 \times 8$ mm, for x , y , z , respectively.

To measure the diffusion and restriction effects, the diffusion coefficient was estimated for a number of diffusion times, which were varied by increasing the TM period. For each diffusion time, the diffusion gradients were stepped linearly, resulting in a number of spectra with increasing diffusion attenuation. This was repeated with the gradient in three different directions (g_x , g_y , and g_z).

The experiments were carried out on a horizontal bore 4.7 T GE CSI instrument, with shielded gradients. Each diffusion measurement used 17 gradient amplitudes, up to 160 mT/m for the shorter diffusion times. The gradient time (δ) was always 10 ms, and the TM period was chosen as 50, 100, 200, 400, and 800 ms. The gradient separation (Δ) was

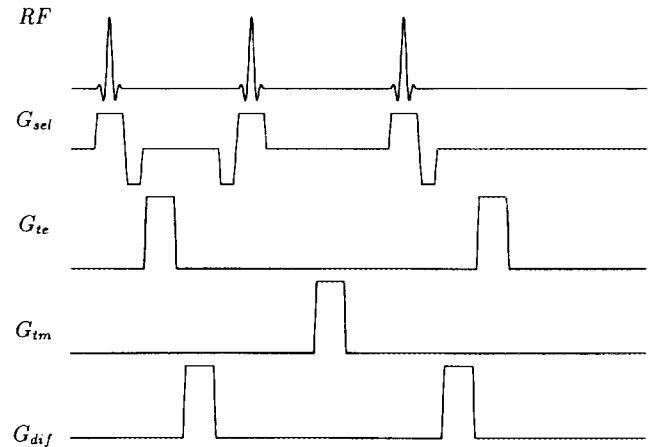


FIG. 1. The stimulated-echo sequence.

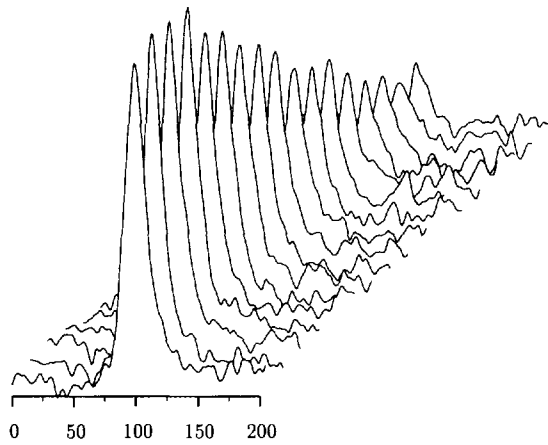


FIG. 2. The ^{31}P spectrum of PCr in rabbit leg muscle as function of the diffusion-gradient amplitude, for TM 400 ms and gradients ranging from 0 to 80 mT/m.

equal to TM plus 12.8 ms. For each gradient value, four averages were acquired, except for TM 800 ms, where the increased T_1 attenuation made eight repetitions necessary. The TE and TM crushers were always chosen in directions perpendicular to the diffusion gradients, to avoid cross terms in the diffusional attenuation.

The spectra were phased and the absorption signal was integrated after a baseline correction. The phasing turned out to be constant for different gradient amplitudes, which indicates that there are no motion artifacts or gradient mismatches in these experiments. The resulting intensities were used to fit an exponential decay, resulting in an amplitude and a diffusion coefficient. Note that, despite the restriction effects, the diffusional attenuation is still a single exponent as a function of the gradient amplitude, Eq. [12]. The lower two gradient amplitudes were always discarded. The trace of the diffusion tensor was calculated as pointed out above. The dependence of this value on the diffusion time was then used to fit the expression for the trace with restricted displacement, combining Eq. [9] and Eq. [12], resulting in the unbounded diffusion coefficient and the cell radius.

Three rabbits were used in this study. The animals were sedated by 1% isoflurane, in a $\text{N}_2\text{O}/\text{O}_2$ mixture (7:3) used for the mechanical ventilation. Catheters were placed in the femoral artery and vein, for blood pressure and heart rate monitoring and for infusion, to which 0.8% succinylcholine was added as muscle relaxant. The other hind leg was placed above a horizontal surface coil, diameter 35 mm, with the gluteus muscle on the coil. The coil and the leg were mounted on a rigid support, fitting tightly in the magnet bore. The temperature was stabilized with a water blanket and monitored with a thermocouple throughout the experiments.

RESULTS

As illustration of the diffusion measurements, a series of spectra is plotted in Fig. 2, showing an increasing attenuation with increasing gradient amplitude.

The resulting diffusion coefficients and mean-square displacements are plotted in Fig. 3, for the different directions and as function of the diffusion time. It is clear that the diffusion is anisotropic and depends on the diffusion time. The diffusion for the longer TM periods does not reflect the restriction size, as can be concluded from the plot of the mean-square displacement Fig. 3b, showing that the displacement continues to increase, indicating that the gradient frame did not correspond closely with the fiber cylinder frame. Some orientation effects can nevertheless be noticed given the fact that the displacement in the Y direction appears to start leveling off at longer diffusion times. This can be explained by the orientation of the muscle fibers, which were predominantly in the XZ plane. Other evidence of orientational effects is given in Fig. 4, which is a plot of the diffusion coefficient as function of the gradient direction in the XZ plane. As pointed out under Theory, the summed dif-

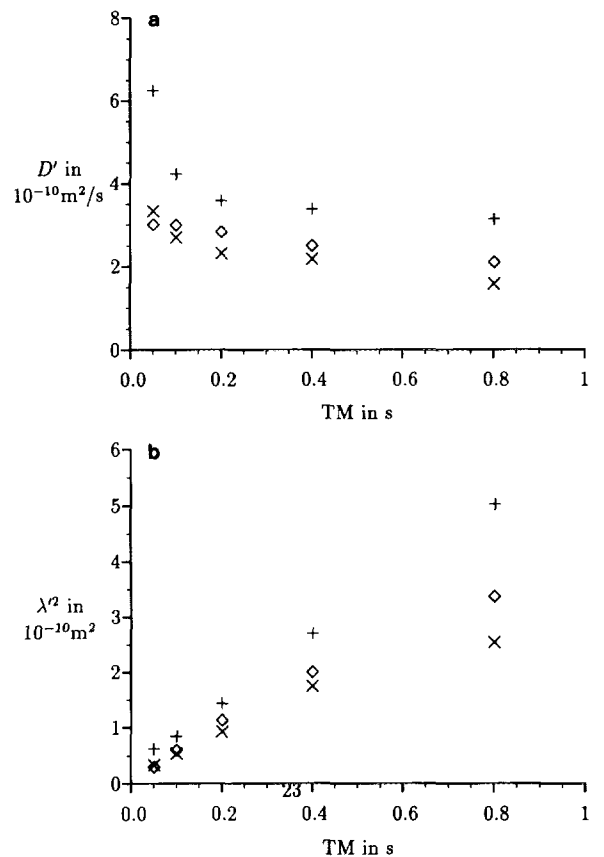


FIG. 3. The diffusion coefficient (D') and the mean-square displacement (λ^2) in the gradient frame of reference, as a function of TM time in direction x (+), direction y ($\times$), and direction z ($\diamond$). (a) The diffusion coefficient. (b) The displacement.

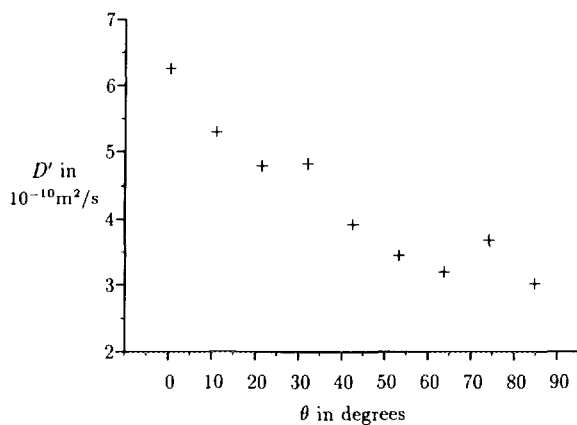


FIG. 4. The diffusion coefficient (D' , gradient frame) as a function of the angle between the gradient and the X axis for TM 100 ms.

fusion in the three orthogonal directions results in the trace and is independent of orientation. This is plotted in Fig. 5a with the solid line showing the fit to Eq. [12]. The corresponding mean-square displacement is plotted in Fig. 5b. The resulting parameters from this fit are printed in Table 1. The measurements of all the three animals resulted in very similar plots of diffusion attenuation and similar values for the cell diameter and intracellular diffusion coefficients (Table 1).

From the residue after the data were fitted, the accuracy in the measurement appeared to be about 1.5%. This leads to an accuracy in the individual diffusion coefficients of approximately 2%. For increasing TM time, the accuracy goes down to approximately 4% because of the decrease in signal intensity. The data of the second animal showed larger deviations; the estimated accuracy of the diffusion in these measurements is about 4–8% (depending on the TM time).

CONCLUSIONS

The orientation effects appear to be of crucial importance for the correct interpretation of the restriction. Without knowledge of the orientation of the cells in the volume of interest, it is not possible to measure the restriction size and/or the unbounded diffusion coefficient using a single gradient direction. The combination of measuring the diffusion in different directions and calculating the trace of the laboratory-frame diffusion tensor is a good and simple solution to this problem.

It is shown that ^{31}P NMR diffusion spectroscopy on rat muscle is a reasonable method for assessing the mobility of phosphocreatine (PCr) under *in vivo* conditions. It also results in an estimate of the cell dimensions, in agreement with literature data obtained with invasive methods (10–100 μm) (12) and data from proton NMR diffusion experiments on excised tissues (8 μm for heart tissue) (13). It should be noted that if the cell diameter is known with sufficient ac-

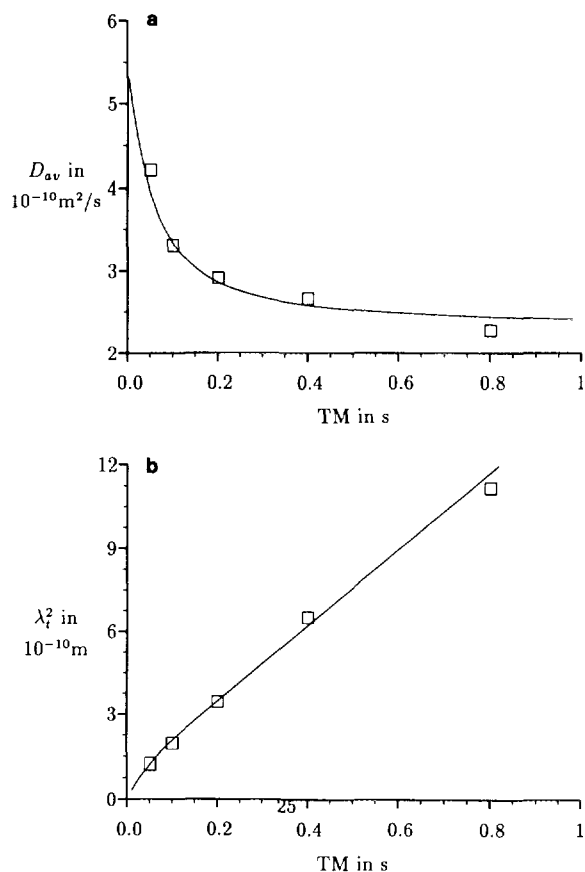


FIG. 5. The average diffusion coefficient [$D_{av} = \frac{1}{3}\text{Tr}(\mathbf{D})$] and the mean-square displacement ($\lambda_r^2 = \lambda_x^2 + \lambda_y^2 + \lambda_z^2$) as a function of length of TM. (a) The diffusion coefficient. (b) The displacement.

curacy, the diffusion coefficient could be measured more precisely.

The diffusion coefficient of phosphocreatine in rabbit muscle is close to that found for phosphocreatine in water at the same temperature (4). This demonstrates the very high diffusional mobility of phosphocreatine in the living muscle. PCr is important for the energy metabolism of muscle cells, although its precise role is still not completely understood. It is thought to function as an energy reservoir.

TABLE 1

Calculated Unbounded Diffusion Coefficient D_f of PCr in Rabbit Leg Muscle Including the Radius R of the Muscle Fibers for Three Animals

Animal	R (μm)	D_f ($10^{-10} \text{ m}^2/\text{s}$)
1	8.9	6.9
2	7.8	9.2
3	9.2	7.1

Together with ATP, it is also essential for the transport of energy, because the location of the oxidative metabolism is different from the location where the generated energy is used. Unfortunately, the fast transverse relaxation rate of the phosphorus resonances of ATP precluded *in vivo* determinations of its diffusion coefficient with the currently available gradients. Nevertheless, because the *in vivo* diffusion coefficient of phosphocreatine is already higher than that of ATP in water at 37°C, the results reported here suggest that phosphocreatine is a more efficient energy transporter in the cytosol than ATP.

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